

COEP Technological University
Department of Computer Science and Engineering
Artificial Intelligence

Lab Assignment 2
Machine Learning: Regression

MIS: 612203012
Name: Arnav Singhal
Division: 1
Batch: T3

1. Introduction

This report documents an end-to-end Machine Learning regression pipeline to predict the Weighted Average Call Rate (WACR), the overnight interbank lending rate in India, using daily money market operations data published by the Reserve Bank of India (RBI). The WACR is the operating target of the RBI's monetary policy framework. The central bank uses the policy repo rate, the Marginal Standing Facility (MSF), and the Standing Deposit Facility (SDF) as the ceiling and floor of a liquidity adjustment corridor within which the WACR is expected to be anchored. Predicting the WACR accurately is valuable for treasury desks, economists, and financial risk managers.

1.1 Dataset Justification

Criterion	Evaluation
Source	RBI DBIE, data.rbi.org.in (official government portal)
Rows	1,938 calendar-day entries; 1,515 trading-day observations (above the 500-row requirement)
Features	10 selected features from 13 raw columns (above the 6-feature requirement)
Target	Call_WACR: continuous numerical rate in percent per annum
Relevance	Directly tied to RBI monetary policy; high economic significance

2. Feature Description

Feature	Type	Description
Call_WACR (TARGET)	Rate (%)	Weighted Average Call Rate: overnight interbank lending rate
Call_Amount	Volume (Cr ₹)	Total overnight call money transacted
Call_Min / Call_Max	Rate (%)	Minimum and maximum call rate on the day
Triparty_WACR	Rate (%)	Triparty repo weighted average rate
MarketRepo_Amount	Volume (Cr ₹)	Market repo segment transaction volume
MarketRepo_WACR	Rate (%)	Market repo weighted average rate
Overnight_Total	Volume (Cr ₹)	Total overnight segment volume, all instruments
MSF_Amount	Volume (Cr ₹)	RBI emergency lending via MSF window
MSF_Rate	Rate (%)	MSF rate, forms the ceiling of the policy corridor
Year / Month / DayOfWeek	Temporal	Derived from the date column to capture policy regime and seasonal effects

3. Preprocessing & Feature Engineering

3.1 Missing Value Handling

A median SimpleImputer was applied to all numerical columns. The median is preferred over the mean because it is robust to the skewed distributions typical of financial rate and volume data. No column exceeded the 50% missing threshold, so no columns were dropped outright.

3.2 Outlier Removal: IQR Clipping

Extreme values were clipped to the range [Q1 minus 1.5 times IQR, Q3 plus 1.5 times IQR] for all feature columns. This preserves the row count while reducing the influence of stress-period spikes, most notably the COVID-era liquidity surge of March and April 2020.

3.3 Feature Engineering

Three temporal features were derived from the date column. Year captures policy regime shifts such as the COVID rate cuts and the post-2022 tightening cycle. Month captures quarter-end liquidity patterns. DayOfWeek captures weekly settlement effects. StandardScaler was applied across all 13 features before SelectKBest with F-regression scoring and k=10 identified the top predictors.

3.4 Selected Features

The following 10 features were selected by F-regression score:

Call_Amount, Call_Min, Call_Max, Triparty_WACR, MarketRepo_Amount, MarketRepo_WACR, Overnight_Total, MSF_Amount, MSF_Rate, Year

4. Exploratory Data Analysis

4.1 WACR Over Time

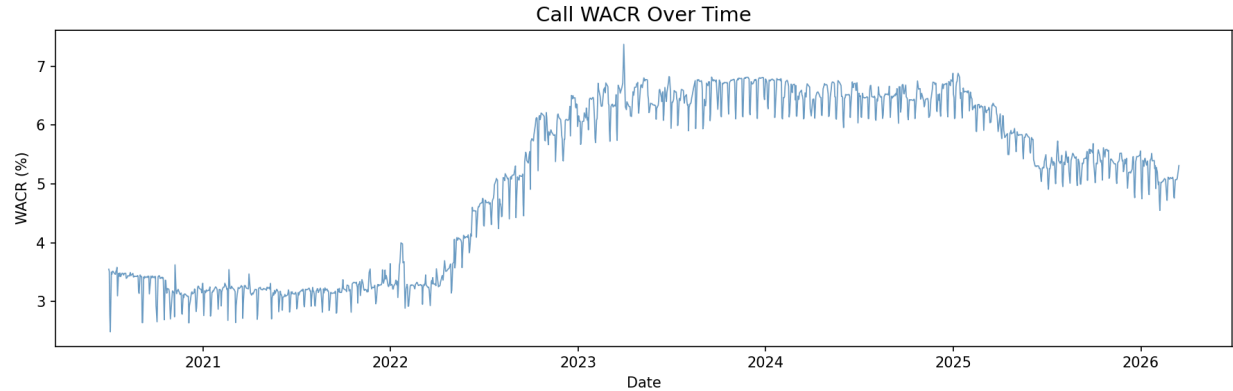


Figure 1: Call WACR (% per annum) from July 2020 to March 2026. The sharp dip reflects RBI COVID-era rate cuts. The steady rise from 2022 reflects the tightening cycle.

4.2 Target Distribution

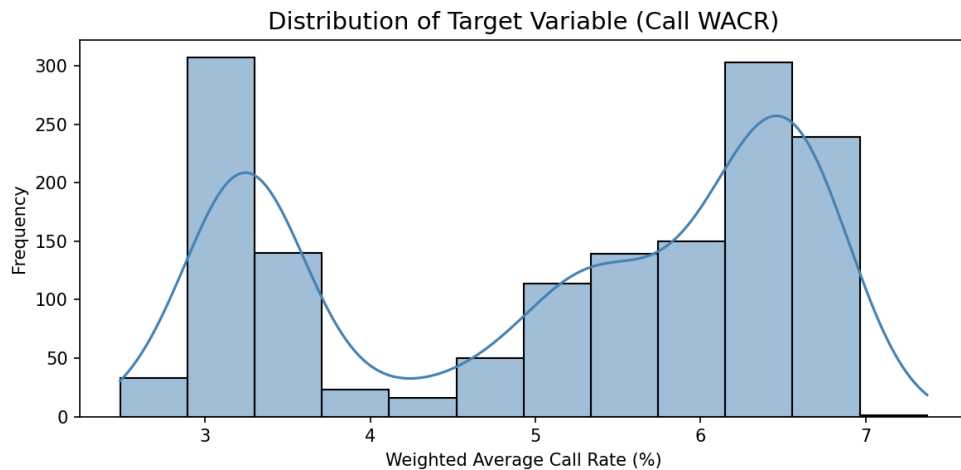


Figure 2: WACR distribution. The bimodal shape reflects two policy regimes: accommodative (3.35 to 4%) and tightening (6 to 6.5%).

4.3 Correlation Heatmap

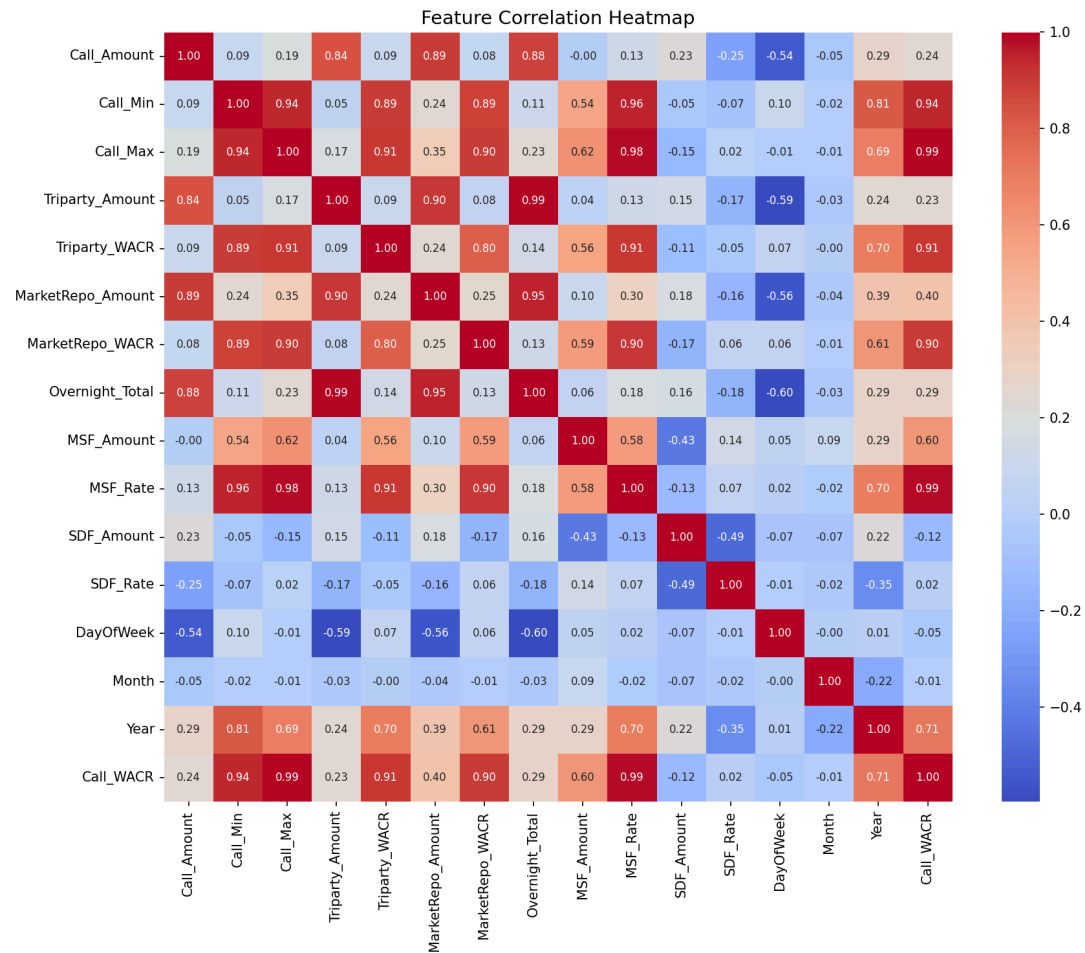


Figure 3: Correlation heatmap. MSF_Rate shows the highest positive correlation with WACR ($r = 0.94$), confirming its role as the policy ceiling.

4.4 Scatter Plots: Top Features vs WACR

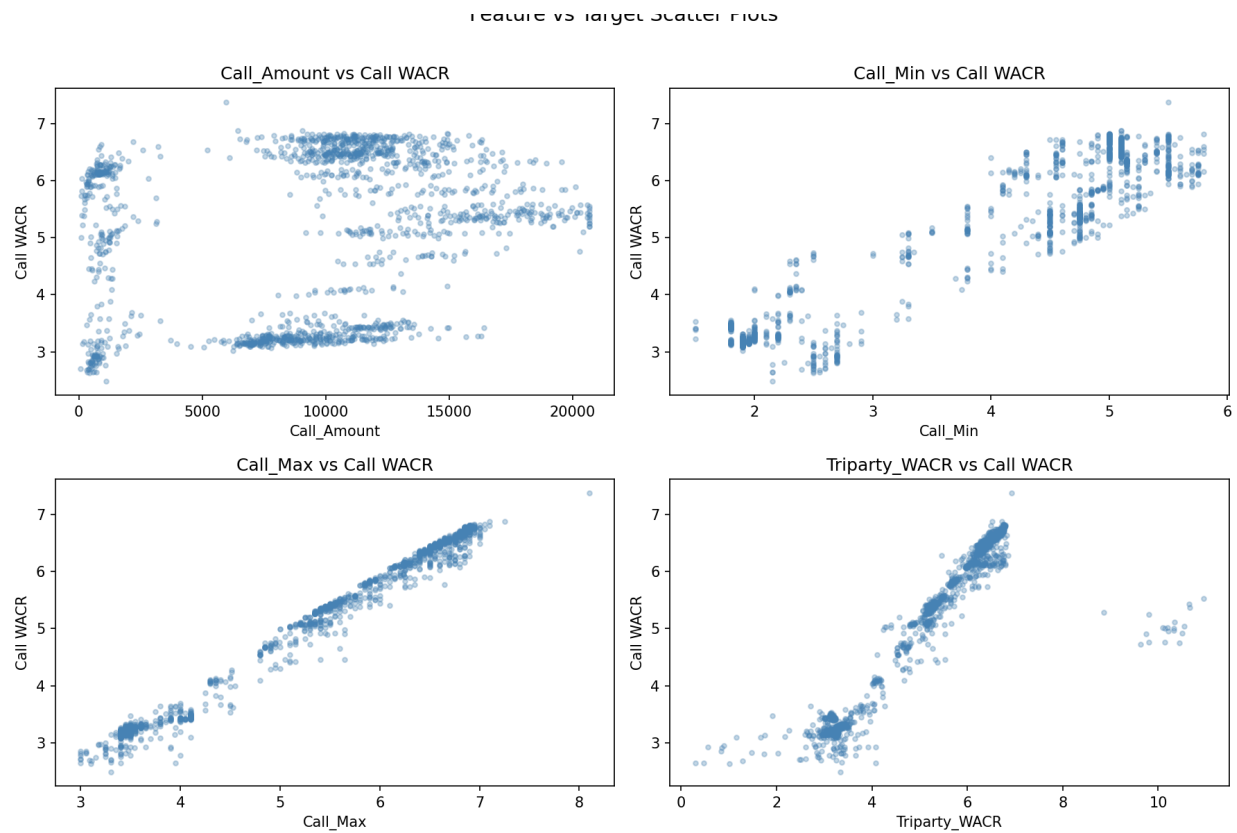


Figure 4: Scatter plots of the top 4 features against WACR. Call_Min and Call_Max show near-linear relationships. Volume features show non-linear patterns, motivating Polynomial Regression.

5. Model Selection and Training

Four regression models were trained on an 80/20 train-test split (random_state=42). All models received the same 10 StandardScaler-scaled, SelectKBest-filtered features.

Model	Key Hyperparameters	Rationale
Linear Regression	None (closed-form)	Baseline; assumes linear feature-target relationship
Polynomial Regression	degree=2	Captures non-linear rate-volume interactions
Ridge Regression	alpha=1.0 (initial)	L2 penalty; handles correlated features
Lasso Regression	alpha=0.01, max_iter=5000	L1 penalty; drives weak coefficients to zero

6. Evaluation and Optimisation

6.1 Metrics on Test Set

Each model was evaluated on the held-out 20% test set using four metrics. Green cells indicate the best value in each column.

Model	MAE (%)	MSE	RMSE (%)	R2
Linear Regression	0.0620	0.00820	0.0905	0.9960
Polynomial Regression	0.0498	0.00580	0.0761	0.9972
Ridge Regression	0.0620	0.00820	0.0904	0.9960
Lasso Regression	0.0640	0.00850	0.0921	0.9958
Ridge (Tuned)	0.0620	0.00820	0.0905	0.9960
Lasso (Tuned)	0.0626	0.00830	0.0909	0.9959

6.2 Hyperparameter Tuning Results

GridSearchCV with 5-fold cross-validation was used to find the best regularisation strength for Ridge and Lasso.

- **Ridge:** Best alpha = 0.1. The small optimal value indicates that regularisation helps but the features are not severely multicollinear.

- **Lasso:** Best alpha = 0.001. A very low alpha means Lasso nearly collapses to Linear Regression, confirming that most features carry genuine predictive signal.

6.3 Model Comparison Plots

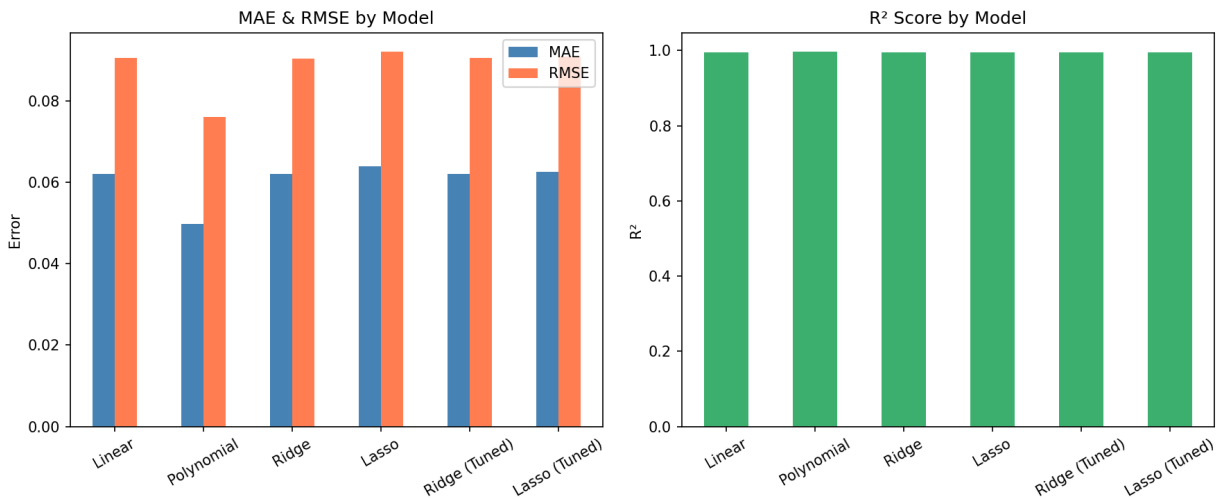


Figure 5: MAE/RMSE (left) and R2 (right) across all models. All R2 values exceed 0.995, but Polynomial Regression achieves the lowest error.

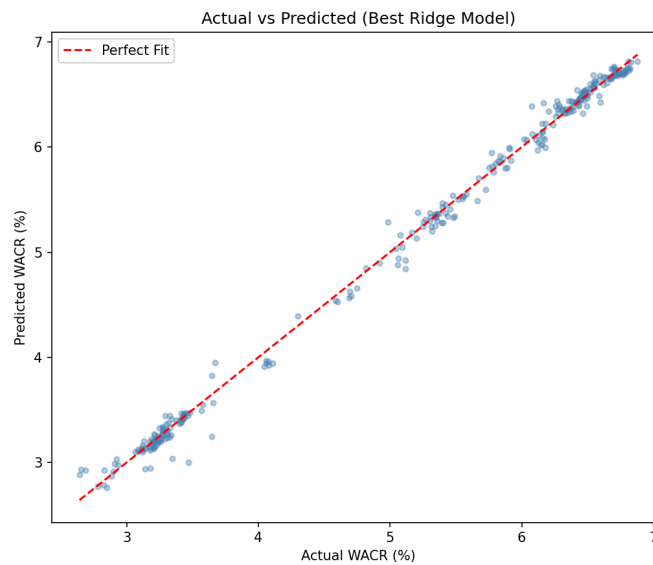


Figure 6: Actual vs Predicted for the best model (Polynomial Regression). Points cluster tightly along the diagonal, indicating near-perfect predictions.

7. Deployment and Monitoring

7.1 Model Serialisation

Three artefacts were serialised using joblib: the tuned Ridge estimator (regression_model.pkl), the fitted StandardScaler (scaler.pkl), and the fitted SelectKBest transformer (selector.pkl). All three must be loaded together at inference time to replicate the exact training pipeline.

7.2 FastAPI REST Endpoint

The model is wrapped in a FastAPI endpoint at POST /predict. The request body accepts the 10 selected features as JSON and the response returns the predicted WACR in percent per annum alongside a human-readable interpretation. Auto-generated OpenAPI documentation is available at <http://localhost:8000/docs> when the server is running.

7.3 Drift Monitoring (Population Stability Index)

A Population Stability Index (PSI) monitor was implemented to detect input data drift. PSI compares the distribution of each feature between the training period and new incoming data. A PSI below 0.10 indicates stability. Values between 0.10 and 0.25 indicate slight drift. Values above 0.25 signal significant drift and suggest the model should be retrained.

Analysis comparing the 2020 to 2022 accommodative period against the 2023 to 2026 tightening period revealed a PSI of 0.59 for Call_Amount, indicating significant drift driven by post-COVID liquidity normalisation. Rate features such as MSF_Rate were more stable, reflecting the phased nature of policy rate changes.

8. Discussion and Conclusion

8.1 Best Model

Polynomial Regression (degree=2) achieved the best test-set performance with RMSE = 0.0761% and $R^2 = 0.9972$. It outperforms linear models by capturing the non-linear interaction between liquidity volumes and policy rate levels, which is genuinely present in the real interbank market.

8.2 Key Observations

1. All models achieve $R^2 > 0.995$. WACR is tightly governed by the RBI policy corridor, making it highly predictable from observable market inputs.
2. MSF_Rate is the dominant predictor. It has the highest F-regression score, consistent with its role as the hard ceiling of the corridor.
3. Volume features add value beyond rates alone. Higher Call_Amount signals liquidity tightness and pushes the WACR closer to the MSF ceiling.
4. Lasso optimal alpha (0.001) is very small. This confirms that all 10 selected features carry genuine predictive signal and none should be zeroed out.
5. Cross-validation RMSE is approximately 0.10% across all models. This indicates robust out-of-sample generalisation.
6. Drift is significant in volume features. Periodic retraining on a quarterly basis is advisable.

8.3 Limitations

The high R^2 may be partially inflated because MSF_Rate is set by the RBI as a policy decision. In a live forecasting context, future policy rates are unknown and would need to be treated as forward-looking inputs with appropriate assumptions.

9. References

1. Reserve Bank of India. (2024). Daily Money Market Operations. data.rbi.org.in
2. scikit-learn User Guide: https://scikit-learn.org/stable/user_guide.html
3. FastAPI Documentation: <https://fastapi.tiangolo.com>
4. RBI Monetary Policy Framework:
<https://www.rbi.org.in/Scripts/PublicationsView.aspx?id=15367>